

Load Interconnection Request System Impact Study Report

For Queue Project Q20-1599

Mercury 138 kV
February 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1599. The Interconnection Customer proposes a 161.2 MW large load (data center) facility interconnecting at the Mercury 138 kV substation in the Pennsylvania area. The request was received on October 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 532 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,592,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Jobos Tc - Burle 138 kV
Contingency: Bixby Hill - Shayland 345 kV line
Loading: 104.5% (Rating: 197 MVA)
DFAX: 8.3%
Pre-project loading: 97.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jobos Tc - Burle 138 kV	Reconductor line	\$2,592,000	100% (\$2,592,000)
	Total allocated to project		\$2,592,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.